

Week 5 Plan Explanation – Transaction Costs & Robustness

Week Goal

Understand real-world trading frictions and test strategy robustness.

By the end of the week, the intern can quantify trading frictions, test momentum across periods, sectors, parameters, and rebalance schedules, and make an evidence-based robustness judgment.

Task 1: Transaction Cost Models

This task makes trading friction explicit by linking portfolio weight changes to turnover and implementable cost assumptions.

Detailed Work

- Types of costs:
 - Explicit: commissions, fees, taxes
 - Implicit: bid-ask spread, market impact, slippage
- Simple cost model: flat cost per trade (e.g., 5 bps or 10 bps one-way)
- Proportional cost model: cost as a function of trade size and volume
- Turnover: sum of absolute weight changes at each rebalance

Concrete Output

- Implement `apply_transaction_costs(returns, weights, cost_bps)` in `src/backtest.py`
- Compute turnover for the momentum strategy from Week 2

Quality Check

Turnover must be computed from actual weight changes with a stated one-way or round-trip cost convention, and costs must be deducted at the correct dates.

Task 2: Momentum with Costs

This task measures how transaction costs change the momentum strategy's realized return, Sharpe ratio, and drawdown profile.

Detailed Work

- Re-run the momentum quintile backtest with transaction costs
- Compare: gross returns vs net returns for each quintile
- Long-short strategy P&L after costs

Concrete Output

- Plot cumulative returns: gross vs net for Q1, Q5, and long-short
- Create a table: annualized return, Sharpe, max DD – before and after costs
- At what cost level does the long-short spread become zero?

Quality Check

Gross and net series must differ only by the documented cost model, and the break-even cost calculation must reconcile with turnover.

Task 3: Robustness: Subperiods & Sectors

This task tests whether the factor result persists through time and across sectors rather than depending on one favorable segment.

Detailed Work

- Split sample into 2–3 subperiods (e.g., 2005–2010, 2010–2015, 2015–2020)
- Does the factor work consistently across subperiods?
- Sector analysis: compute factor returns within each sector
- Is the factor driven by a single sector?

Concrete Output

- Create subperiod performance table
- Plot factor returns by sector
- Test sector-neutral version: rank within sectors instead of across all stocks

Quality Check

Subperiod and sector results must use the same signal definition, with sector-neutral rankings compared on an equal basis to the broad-universe strategy.

Task 4: Robustness: Parameters & Rebalancing

This task evaluates sensitivity to signal windows, portfolio grouping, and rebalancing frequency without selecting parameters after seeing the answer.

Detailed Work

- Sensitivity to lookback period: test 3–1, 6–1, 9–1 momentum in addition to 12–1
- Sensitivity to number of portfolio groups: terciles vs quintiles vs deciles
- Rebalancing frequency: weekly vs monthly vs quarterly

Concrete Output

- Create a parameter sensitivity heatmap (lookback × rebalance frequency → Sharpe)
- Discuss: which parameters are the strategy most sensitive to?

Quality Check

The sensitivity grid must be defined before interpretation, use comparable samples, and report fragile regions instead of highlighting only the best cell.

Task 5: Synthesis & Write-Up

This task combines the cost and robustness evidence into a clear judgment about whether the strategy is durable or fragile.

Detailed Work

- Consolidate all robustness results
- Discuss: is the momentum factor robust or fragile?
- What would you need to see before deploying this as a live strategy?

Concrete Output

- Create a comprehensive robustness summary table
- Write weekly memo: key findings on costs and robustness
- Friday demo

Quality Check

Every conclusion must be traceable to a table or chart, and deployment claims must acknowledge liquidity, market impact, and out-of-sample validation needs.

Week Deliverables

- `notebooks/week5/transaction_costs_robustness.ipynb`
- Robustness summary: tables and charts showing sensitivity analysis
- Weekly memo (1–2 pages)

Recommended Reading

- Frazzini, Israel, & Moskowitz (2012), "Trading Costs of Asset Pricing Anomalies"
- Novy-Marx & Velikov (2016), "A Taxonomy of Anomalies and their Trading Costs"
- *Advances in Financial Machine Learning* (de Prado), Chapter on Backtesting (for pitfall awareness)

Common Pitfalls

- Underestimating transaction costs (especially for small-cap or illiquid stocks)
- Not accounting for market impact (large positions move prices)
- Overfitting by optimizing parameters on the full sample
- Ignoring turnover when comparing strategies

What Success Looks Like

The intern can quantify trading frictions, test momentum across periods, sectors, parameters, and rebalance schedules, and make an evidence-based robustness judgment.